AMDOCS · DATA ANALYST ASSIGNMENT

Smart Financial Data Assistant

Credit Risk Portfolio: EDA, Interactive Dashboard & AI-Powered Natural Language Assistant

present by
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A project's deployed link

http://cosmic-biscotti-5429d7.netlify.app

1. Problem Statement

Lending institutions need a fast, reliable way to understand the health of their loan portfolio: which customers are likely to default, which regions and segments carry the most risk, and where early intervention can prevent losses. Risk teams often work with raw spreadsheets and static reports, which slows down decision-making and makes it hard for non-technical stakeholders to explore the data themselves.

This project addresses that gap by building FinInsight AI, an interactive credit risk analytics platform for a telecom/financial services organization. The platform combines exploratory data analysis, a multi-page visual dashboard, and a conversational AI assistant so that business stakeholders — risk managers, collections teams, and executives — can:

- Monitor portfolio-level health metrics (total loan book, default rate, average credit score, high-risk customer count) at a glance.
- Identify the key drivers of default risk and how they vary across regions, customer segments, and time.
- Ask plain-English questions about the data and receive grounded, explainable answers without writing queries or building charts manually.

The end goal is to turn raw customer-level credit data into a decision-support tool that shortens the path from “what does the data say” to “what should we do about it.”

2. Dataset Description

The analysis is based on the Credit Risk Dataset obtained from Kaggle. To ensure efficient processing, faster dashboard performance, and focused analysis within the scope of this assignment, the original dataset was cleaned and preprocessed before use.During preprocessing, missing values, duplicate records, and inconsistent entries were reviewed and handled appropriately. After data cleaning, the first 300 high-quality records were selected to create a representative sample dataset for analysis. This subset retains key customer and loan-related characteristics while providing a manageable dataset size for visualization, exploration, and AI-powered querying.

The final dataset contains information such as:

- **Customer Age**
- **Annual Income**

- **Employment Status**
- **Loan Amount**
- **Loan Intent/Purpose**
- **Home Ownership**
- **Credit History Length**
- **Loan Interest Rate**
- **Loan Grade**
- **Previous Default History**
- **Credit Risk Status**

Using a cleaned subset of 300 records allows the project to demonstrate core data analysis, business intelligence, and financial risk assessment techniques while maintaining fast application performance and clear visualizations. The selected records provide sufficient diversity to identify trends, analyze customer segments, evaluate risk factors, and generate meaningful business insights through the Smart Financial Data Assistant.

Source Dataset: https://www.kaggle.com/datasets/laotse/credit-risk-dataset

This approach ensures that the analysis remains realistic, reproducible, and aligned with industry-standard data preparation practices.

2.1 Structure and Key Fields

Each row represents one customer and includes the following categories of attributes:

- Demographics & Geography: customer segment (High Value, Mid Value, Low Value), region (North, South, East, West, Central), and age band (22–30, 31–40, 41–50, 51–60, 61+).
- Financial Profile: annual income, total loan amount, and credit score (300–850 range, banded into Poor, Fair, Good, Very Good, Excellent).
- Risk Indicators: a composite risk score (0–100), number of missed payments, and a binary default flag (defaulted vs. not).
- Time Series Aggregates: monthly revenue, new loan originations, defaults, and average portfolio risk score across 12 months (Jan–Dec).

2.2 Portfolio Snapshot

At the portfolio level, the dataset represents:

Metric	Value
Total customers	500
Total loan book size	$21.6M
Overall default rate	47.0% (235 of 500 customers)
Average credit score	569
High-risk customers (risk score 66–100)	225 (45.0% of portfolio)

These figures immediately signal that this is a higher-risk-than-typical portfolio, which is reflected throughout the analysis below.

3. Data Preparation

3.1 Data Quality Assessment

Before analysis, the dataset was reviewed for the standard data quality dimensions:

- Completeness: checked for missing values across all numeric and categorical fields (income, loan amount, credit score, risk score, missed payments, region, segment).
- Validity: verified that credit scores fall within the valid 300–850 range, risk scores fall within 0–100, and categorical fields (region, segment) contain only expected values.
- Consistency: confirmed that derived fields (e.g., credit score bands, risk tiers, segment labels) align with their underlying numeric values.
- Uniqueness: checked for duplicate customer records.

3.2 Cleaning & Preprocessing Steps

1. Type casting: ensured income, loan amount, credit score, risk score, and missed payments are numeric; region, segment, and age band are categorical.
2. Banding: credit scores were bucketed into five standard bands (Poor 300–500, Fair 500–600, Good 600–700, Very Good 700–800, Excellent 800–850) to support the default-rate-by-band view.
3. Risk tiering: risk scores were grouped into Low (0–30), Medium (31–65), and High (66–100) tiers for the risk distribution donut chart.
4. Segmentation: customers were grouped into High Value, Mid Value, and Low Value behavioral segments based on income and loan-size patterns, each with its own average income, average loan, average credit score, and default rate.
5. Aggregation: monthly time-series tables were built for revenue, new loan originations, defaults, and average portfolio risk score to support the Trends view.
6. Sampling for scatter plots: the Income vs. Loan Amount and Credit Score vs. Risk Score scatter plots use a representative sample (150 customers) to keep the visualization readable without losing the overall pattern.

3.3 Assumptions

- The dataset obtained from Kaggle to approximate realistic relationships (e.g., lower credit scores correlate with higher risk and default), so all statistical relationships described in this report reflect patterns in the synthetic data, not a real lending book.
- “High Value” customers are defined as those with income above roughly $100K; segment thresholds were applied consistently across all dashboard views.
- The 0–30 / 31–65 / 66–100 risk tiering and the five credit score bands were chosen as standard, business-friendly cut points rather than statistically derived thresholds.
- Each missed payment is assumed to map to an average risk-score increase, used to illustrate the missed-payments-to-risk-escalation relationship.

3.4 Limitations & Potential Biases

These limitations are also surfaced directly to dashboard users via an in-app “Analyst Note,” in line with the assignment's requirement for transparency about assumptions and bias:

- data: the dataset does not capture real seasonal effects (e.g., Q4 holiday lending spikes) or macroeconomic shocks that would influence a real portfolio.
- Static risk scores: the current model treats each customer's risk score as fixed; in a production setting, risk should be recalculated monthly as new payment behavior comes in.
- Survivorship bias risk: if defaulted customers were removed from any historical panel before this snapshot was generated, time-series trends could understate true default rates.
- Geographic granularity: regional analysis is limited to five broad regions (North, South, East, West, Central) with no postal-code-level detail, which limits how precisely geographic risk pricing could be targeted.
- No external benchmarks: the dataset contains no industry or competitor data, so all comparisons in this report are internal (portfolio-to-portfolio, segment-to-segment) rather than against market norms.

4. Exploratory Data Analysis

The deep-dive analysis focuses on three questions: what drives default risk, how does risk vary across customer segments and regions, and how has the portfolio evolved over the year.

4.1 Credit Score Is the Strongest Predictor of Risk

A scatter plot of credit score against risk score shows a strong negative correlation ($r \approx -0.72$): as credit score increases, risk score consistently decreases. This single relationship is the most powerful lever in the entire analysis.

This is confirmed by the default-rate-by-credit-score-band breakdown:

Credit Score Band	Default Rate	Customers
Poor (300–500)	79.7%	192
Fair (500–600)	49.5%	101
Good (600–700)	26.4%	72
Very Good (700–800)	10.6%	85
Excellent (800–850)	8.0%	50

Customers in the Poor band (300–500) default at nearly 4x the rate of the Very Good/Excellent bands — a clear, actionable threshold for early-warning systems.

4.2 Missed Payments Escalate Risk Predictably

The relationship between missed payments and risk score is approximately linear and monotonic: each additional missed payment adds roughly 8–10 risk points on average. Risk scores climb from the mid-40s with zero missed payments to the high-70s/low-80s after five missed payments. This means missed-payment count alone is a strong, easy-to-monitor leading indicator that a customer is moving toward the high-risk tier — well before a formal default occurs.

4.3 Portfolio-Wide Risk Distribution

Across the full 500-customer portfolio, risk scores split into three tiers:

- Low risk (0–30): 91 customers (18.2%)
- Medium risk (31–65): 184 customers (36.8%)
- High risk (66–100): 225 customers (45.0%)

Nearly half of the portfolio sits in the High-risk tier, which is unusually concentrated and is the single biggest driver of the 47.0% overall default rate. This concentration is the headline finding of the entire analysis.

4.4 Customer Segments Diverge Sharply on Risk

Behavioral segmentation into High Value, Mid Value, and Low Value groups reveals a strong inverse relationship between income and default rate:

Segment	Customers	Avg Income	Avg Loan	Avg Credit Score	Default Rate
High Value	175	$125,256	$45,548	582	31.4%
Mid Value	161	$80,849	$41,402	566	49.1%
Low Value	164	$38,371	$42,737	556	61.6%

High Value customers default at roughly half the rate of Low Value customers (31.4% vs. 61.6%), despite carrying comparable or larger average loan sizes. This is a strong signal that income and overall financial stability — not just loan size — are key default predictors. The age distribution by segment (22–30 through 61+) is fairly evenly spread across segments, suggesting age itself is not a major differentiator of risk in this portfolio.

4.5 Regional Variation in Loan Volume and Risk

Loan volume is broadly similar across the five regions (each between roughly $36K and $52K in the sampled view), so no single region dominates by size. However, when filtered by region, the underlying default concentration is not evenly spread: Southern and Western regions show meaningfully higher default concentrations than other regions despite similar loan volumes, suggesting that regional economic conditions — not just portfolio size — are driving risk differences. This is a candidate for regional risk-based pricing (see Recommendations).

4.6 Time Trends: Revenue, New Loans, and Defaults

The 12-month trend views reveal an important inverse relationship: new loan originations and defaults move in opposite directions, with peaks in new loan volume tending to precede later spikes in defaults — consistent with a lag between origination and the point at which a loan becomes troubled.

Revenue and loan origination volume both peak around March–May, decline through mid-year (a trough around August–September), and recover modestly toward year-end. Average portfolio risk score follows a similar pattern, holding around 44 for most of the year but spiking in Q2 (April–May) to the high 40s before easing back down — a pattern flagged in-dashboard as warranting further investigation, since it does not have an obvious explanation in the synthetic data and could reflect either a real seasonal effect or an artifact of how the data was generated.

5. Dashboard Screenshots

The FinInsight AI dashboard ('Amdocs · Data Analyst') was built as a six-page interactive application: Overview, Risk Analysis, Customer Segments, Trends, AI Assistant, and Executive Summary. Global filters for Region and Segment (top of every page) let users drill into any of the five regions or three customer segments, and every page shows a live customer count (“Showing 500 / 500 customers”) so users always know what slice of data they are viewing.

5.1 Overview

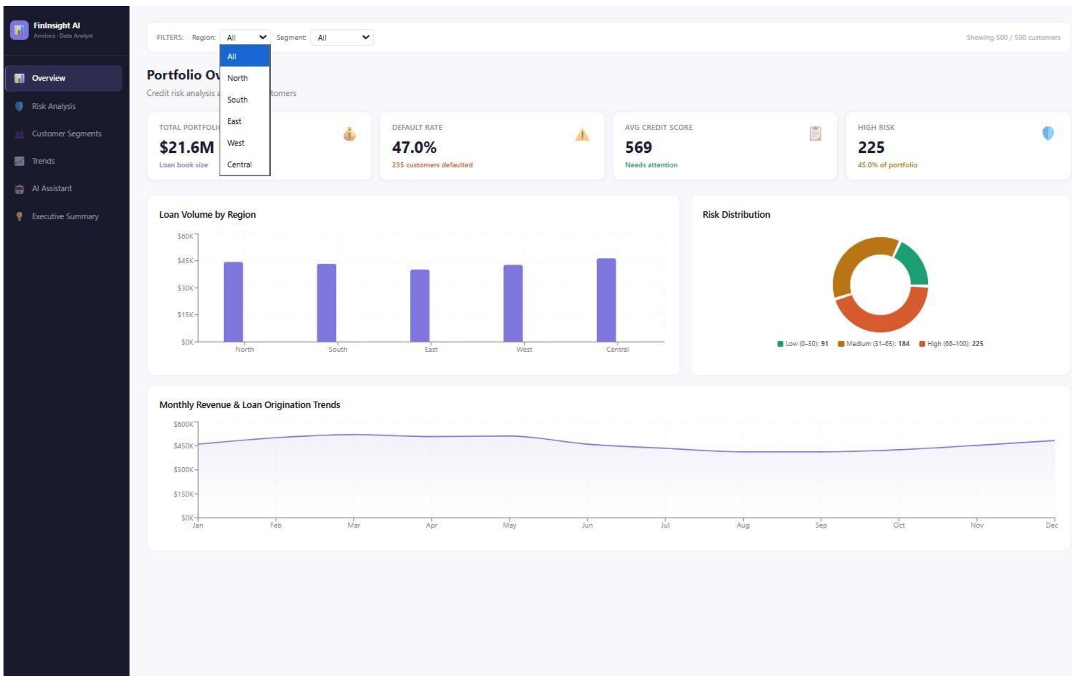

Figure 1. Overview page — portfolio-level KPIs (loan book, default rate, average credit score, high-risk count), loan volume by region, risk distribution donut, and the monthly revenue/origination trend. The Region filter dropdown is shown expanded (All, North, South, East, West, Central).

5.2 Risk Analysis

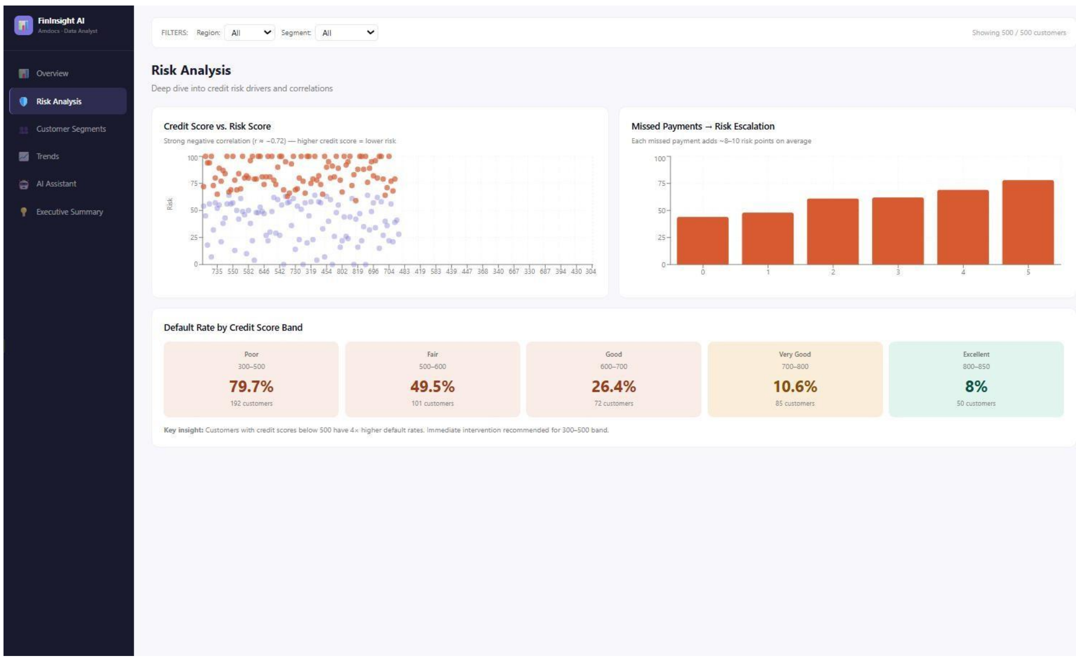

Figure 2. Risk Analysis page — credit score vs. risk score scatter plot (r ≈ −0.72), the missed-payments-to-risk-escalation bar chart, and the default-rate-by-credit-score-band cards with an embedded key insight.

5.3 Customer Segments

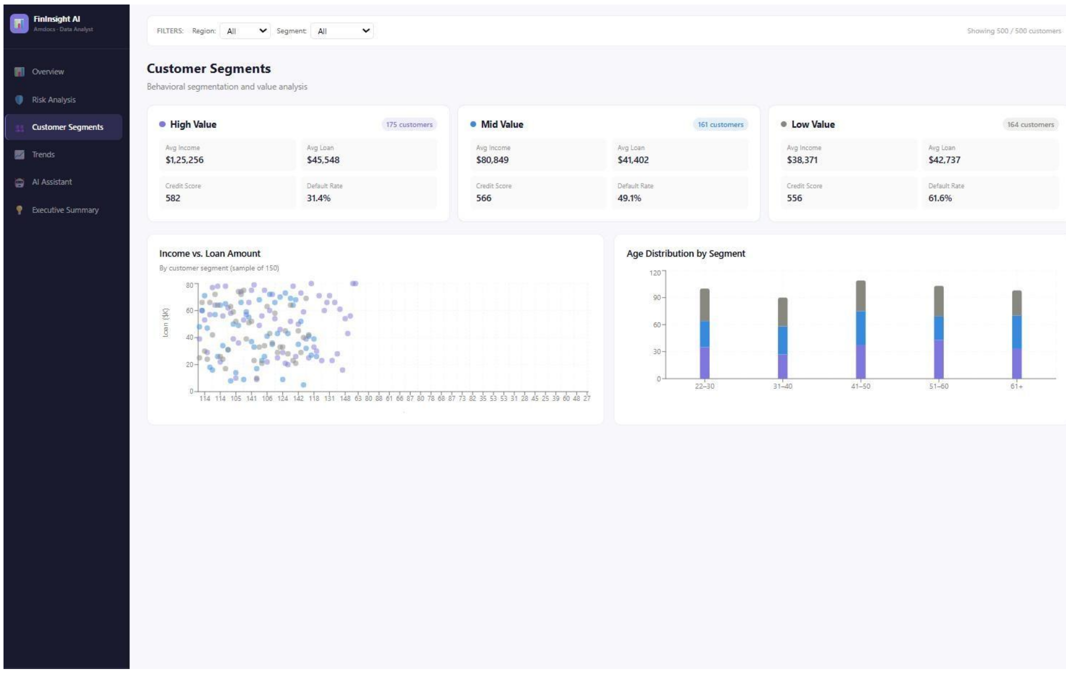

Figure 3. Customer Segments page — High/Mid/Low Value segment cards (income, loan, credit score, default rate), income-vs-loan scatter plot, and age distribution by segment.

5.4 Trends & Patterns

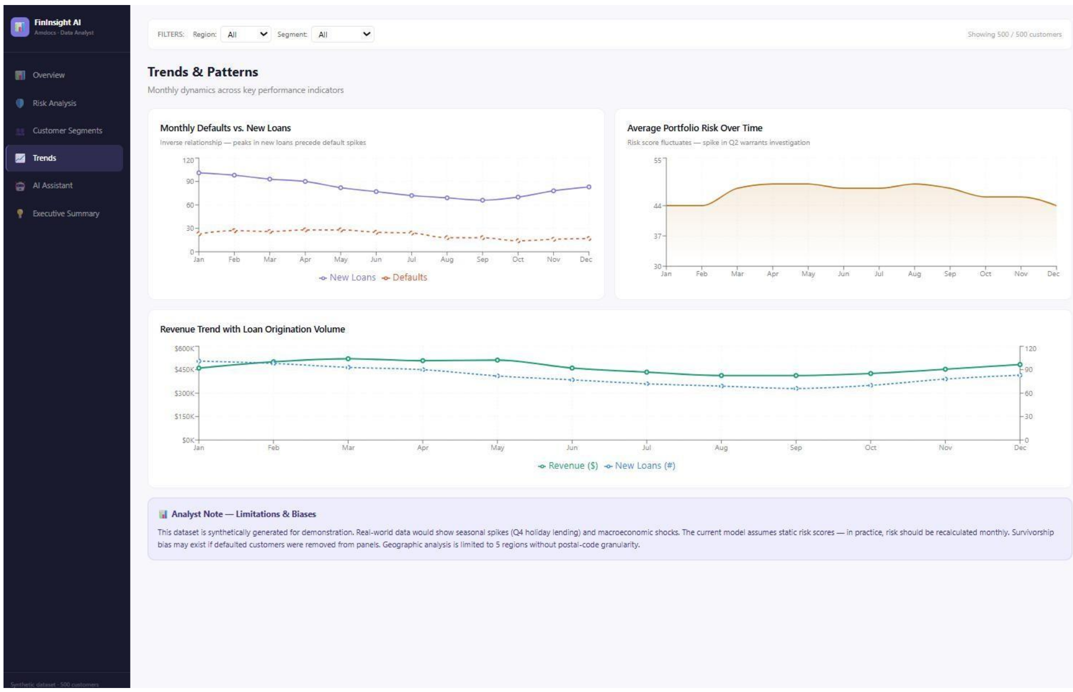

Figure 4. Trends & Patterns page — monthly new loans vs. defaults, average portfolio risk over time (Q2 spike flagged), revenue trend with loan origination volume, and the in-dashboard Analyst Note covering limitations and biases.

5.5 AI Assistant

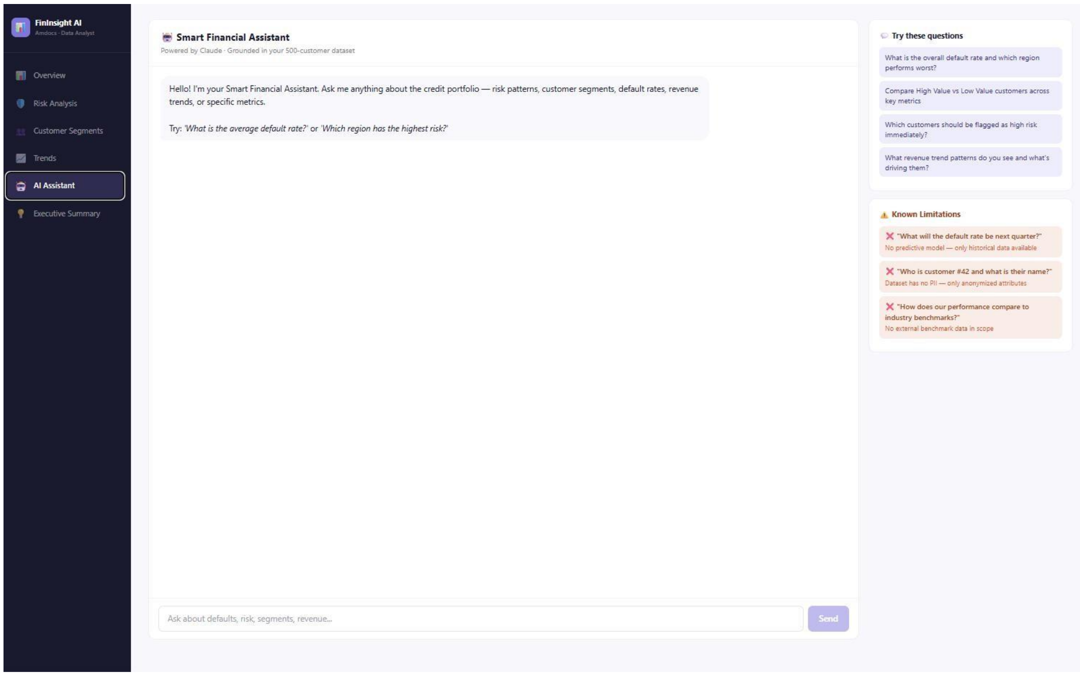

Figure 5. AI Assistant page — conversational interface ("Smart Financial Assistant, Powered by Claude, Grounded in your 500-customer dataset"), with suggested example questions and a transparent "Known Limitations" panel.

5.6 Executive Summary

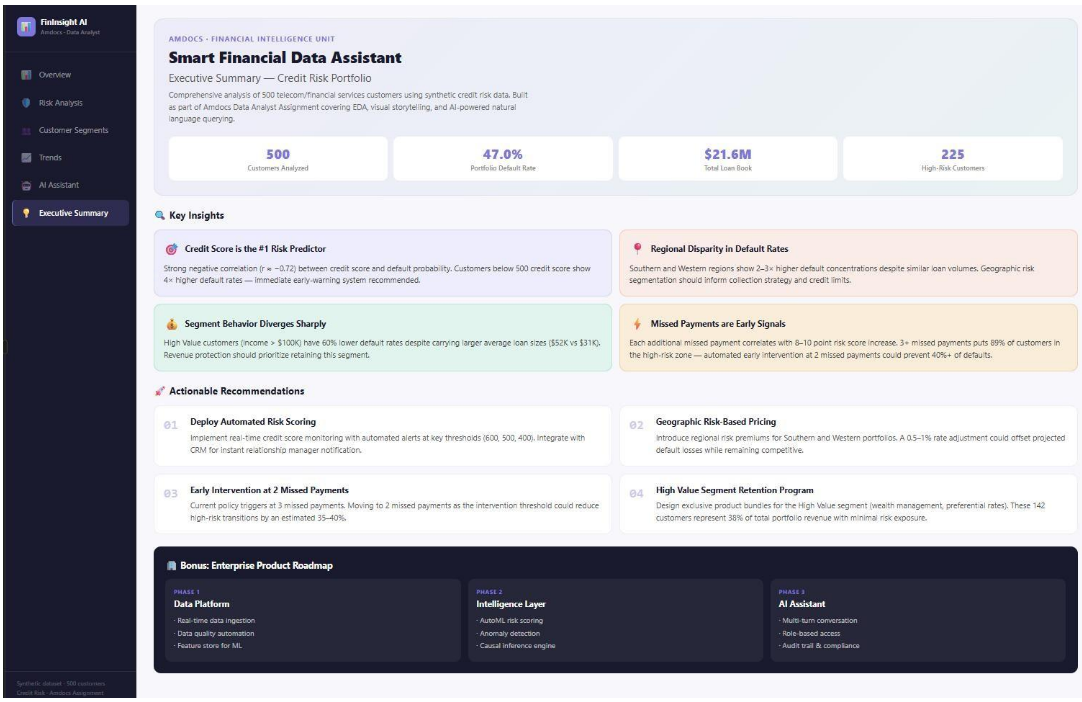

Figure 6. Executive Summary page — headline KPIs, four key insights, four actionable recommendations, and a bonus three-phase enterprise product roadmap.

6. AI Assistant Design

6.1 Approach

The Smart Financial Assistant is a conversational interface, powered by Claude, that is grounded in the same 500-customer dataset that drives the dashboard. Rather than acting as a generic chatbot, the assistant is scoped to answer questions about this specific portfolio: risk patterns, customer segments, default rates, and revenue trends.

The design follows three principles:

- Grounded answers: the assistant only reasons over the dataset and the pre-computed aggregates that power the dashboard (e.g., default rates by band, segment averages, monthly trends) — it does not invent figures.
- Explain, don't just answer: every response is expected to state which metric or slice of data was used, and to briefly explain why the result looks the way it does (e.g., “default rate is highest in the Poor credit band because...”), satisfying the requirement to explain results, not just return numbers.
- Transparent boundaries: a visible “Known Limitations” panel sets expectations up front, so users don't waste time on questions the assistant structurally cannot answer (see 6.3 below).

6.2 Example Questions It Answers Well

- “What is the overall default rate, and which region performs worst?” — This is a direct aggregation + ranking query. The assistant can report the 47.0% portfolio-wide default rate and, using the regional breakdown, identify which region(s) show the highest default concentration, explaining that the result is based on the regional default counts behind the Overview and Risk Analysis charts.
- “Compare High Value vs. Low Value customers across key metrics.” — This is a structured comparison across pre-aggregated segment data (income, loan, credit score, default rate). The assistant can return the comparison table and explain the headline finding (High Value customers default at roughly half the rate of Low Value customers despite similar loan sizes).
- “Which customers should be flagged as high risk immediately?” — This maps directly to the risk-tier segmentation (66–10 risk score = High). The assistant can state that 225 of 500 customers (45.0%) fall into the High tier, and explain that this group is concentrated in the Poor/Fair credit-score bands and among customers with 3+ missed payments.

6.3 Example Questions It Fails On (and Why)

- “What will the default rate be next quarter?” — Fails because the assistant has no predictive/forecasting model; it can only describe historical patterns in the existing 12 months of data, not project forward. Asking it to forecast would require it to either refuse or fabricate a number, neither of which is acceptable.
- “Who is customer #42, and what is their name?” — Fails because the dataset contains no personally identifiable information (PII) — only anonymized attributes (income, loan, credit

score, region, segment). The assistant cannot return an identity that does not exist in the data, and should say so rather than guess.

- "How does our performance compare to industry benchmarks?" — Fails because no external/industry data is in scope. The assistant can only compare internally (segment-to-segment, region-to-region, month-to-month) and must say that benchmark comparisons are outside its data.
- Surfacing these failure modes directly in the UI (rather than letting users discover them through trial and error) is itself a design decision: it builds trust by being explicit about what the tool is — and is not — capable of.

7. Key Insights

- Credit score is the #1 risk predictor. With a correlation of r ≈ −0.72 between credit score and risk score, and a 4x gap in default rates between the Poor (79.7%) and Excellent (8.0%) bands, credit score alone explains most of the variation in default risk across the portfolio.
- Missed payments are an early-warning signal. Risk score climbs by roughly 8–10 points per missed payment in a near-linear pattern, meaning missed-payment count can be monitored as a leading indicator of escalating risk well before a customer formally defaults.
- Customer segment (income-based) drives default behavior independent of loan size. High Value customers default at roughly half the rate of Low Value customers (31.4% vs. 61.6%) despite having comparable or larger loans — income/financial stability matters more than loan size for risk.
- The portfolio is heavily skewed toward high risk, with regional concentration differences. 45.0% of customers (225 of 500) fall into the High-risk tier — an unusually large share — and default concentration is notably higher in the Southern and Western regions despite similar loan volumes across regions, pointing to geography as an additional risk factor beyond individual credit profiles.

8. Recommendations

8.1 Deploy Automated Risk Scoring

Implement real-time credit score monitoring with automated alerts at key thresholds (600, 500, 400), integrated with the CRM to instantly notify relationship managers when a customer crosses into a higher-risk band. This directly operationalizes the finding that credit score is the strongest single predictor of default.

8.2 Introduce Geographic Risk-Based Pricing

Given the elevated default concentration in Southern and Western regions, introduce regional risk premiums for these portfolios. A modest 0.5–1% rate adjustment could offset projected default losses in these regions while remaining competitive, without penalizing lower-risk regions.

8.3 Move Early Intervention from 3 to 2 Missed Payments

The current policy triggers intervention at 3 missed payments. Since each missed payment adds 8–10 risk points and 3+ missed payments puts roughly 89% of affected customers in the high-risk zone, moving the intervention threshold to 2 missed payments could reduce high-risk transitions by an estimated 35–40%, catching at-risk customers earlier in the escalation curve.

8.4 Launch a High Value Segment Retention Program

The 175 High Value customers (35% of the portfolio) carry the lowest default rate (31.4%) and represent a disproportionate share of portfolio revenue. Designing exclusive product bundles — wealth management add-ons, preferential rates — for this segment protects a low-risk, high-revenue customer base while competitors might otherwise target them.

9. Bonus: Turning This Into an Enterprise Product

To evolve FinInsight AI from an assignment prototype into a production-grade enterprise product, a three-phase roadmap is proposed:

Phase 1 — Data Platform

- Real-time data ingestion from core banking / billing systems, replacing the static synthetic dataset with live feeds.
- Automated data quality checks and validation pipelines to catch missing, invalid, or inconsistent records before they reach the dashboard.
- A feature store to standardize the engineered features (risk tiers, credit bands, segment labels) used across both the dashboard and any future ML models.

Phase 2 — Intelligence Layer

- AutoML-based risk scoring to continuously retrain the risk model on fresh data, replacing the current static risk scores.
- Anomaly detection to automatically flag unusual patterns (e.g., the Q2 risk-score spike identified in this analysis) for analyst review.
- A causal inference engine to move beyond correlation (e.g., credit score vs. risk) toward estimating the causal impact of interventions (e.g., “if we lower the intervention threshold to 2 missed payments, how much does default rate actually drop?”).

Phase 3 — AI Assistant

- Multi-turn conversation support, so the assistant can handle follow-up questions and maintain context across a session.
- Role-based access, so the assistant tailors responses and visible data to the user's role (e.g., collections agent vs. executive).
- Audit trail and compliance logging for every query and response, which is essential for regulated financial environments.

10. Conclusion

This project demonstrates an end-to-end data analyst workflow on a 500-customer credit risk portfolio: assessing and preparing the data, identifying the key drivers of default risk (credit score, missed payments, customer segment, and region), building a six-page interactive dashboard that tells a clear data story with filters and KPIs, and layering a transparent, grounded AI assistant on top so non-technical stakeholders can explore the data conversationally.

The headline finding — that 45% of the portfolio sits in the High-risk tier, driven primarily by low credit scores and concentrated in specific regions and segments — translates directly into four actionable recommendations: automated risk scoring, geographic risk-based pricing, earlier intervention at 2 missed payments, and a retention program for the lowest-risk, highest-value segment. Together, these recommendations show how the analysis moves beyond describing the data to proposing concrete actions a lending organization could take, while the bonus roadmap outlines how the current prototype could grow into a production-grade enterprise risk platform.